

PHAM ANH TUAN

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EDUCATION

National University of Singapore | GPA: 5.0/5.0 (Post S/U) Aug 2023 - present
B. Eng. | Robotics and Machine Intelligence (2nd Major: Quantitative Finance, Minor: Innovation and Design) Singapore

Relevant Coursework: Calculus, Differential Equations, Linear Algebra, Fundamentals of Robotics, Fundamentals of Artificial Neural Network, Programming Methodology, Kinematics and Dynamics, Robot Programming

Ly Tu Trong High School for the Gifted | GPA: 9.7/10.0 August 2022 – May 2025
Concentration | Physics Can Tho, Vietnam

Leadership: Class President; Founder & President, Chess Club; Vice President, Mathematics Club

HONORS & AWARDS

- 1st place**, Robosub 2026 Autonomy Challenge, RoboNation July 2026
- 4th place**, RoboRoarZ AI Challenge, Singapore University of Technology and Design January 2026
- 3 Bronze Medals**, Vietnamese Championship for Chess Clubs, Vietnam Chess Federation June 2025
- Bronze Medal**, Vietnam National Physics Olympiad (VPhO), Ministry of Education January 2025

EXPERIENCE

Software Engineering Intern August 2026 - present
BeeX Singapore

- Built a **real-time** feature extraction pipeline utilizing vector voting on disparity maps to accurately identify graspable candidate points on novel objects.
- Integrated **3D point clouds** reconstructed from stereo vision to propose optimal grasping positions and execute reliable trajectory motion planning.

Software Engineer September 2025 - present
Bumblebee Autonomous Systems, National University of Singapore Singapore

- Designed a competition-ready task system using **Py_trees** to build **behavior trees** that handled complex robot decision making.
- Developed a PnP-based pose estimator capable of resolving multi-solution ambiguities, achieving a **95%+** successful localization rate.
- Developed a tracer for **underwater pipeline localization** using 2D sonars, achieving **92%** accuracy.

SELECTED PROJECTS

Learning Continuous LiDAR Navigation Sep 2026

- Designed and trained a **4-stage imitation-to-RL pipeline** — expert, **behaviour cloning**, **DAGger**, **teacher-anchored PPO** — enabling a holonomic robot to cross randomised 7.2 m cone fields from a **36-ray LiDAR scan** alone.
- Diagnosed three **simulation faults presenting as policy failures** — collision threshold, reset race, filtered demonstration labels — recovering expert success from **20% to 85%** and eliminating collisions entirely (14 → 0).
- Achieved **83.7% success (251/300 episodes)**, a **+19.3-point** gain over the DAGger 3 baseline, and validated the frozen checkpoint at **83.0%** on independent re-evaluation.

Image Inpainting April 2026

- Designed & trained **CAE** and **Vision Transformer** architectures for **image inpainting** on **CIFAR-10** & **Oxford-IIIT Pet**, benchmarking optimizer, latent dimensionality, and masking strategies.
- Engineered a **7-phase curriculum training schedule** with **adaptive MAE + perceptual loss weighting**, enabling coherent reconstructions under **69% random occlusion**.
- Validated **cross-domain generalisation** to unseen object categories, confirming the model learned general inpainting priors rather than texture memorisation.

Hospitality AI January 2026

- Implemented a **low-latency, multi-stage ML pipeline** using **ResNet-18** & **YOLOv8n** for real-time bed quality inspection.
- Built **augmentation methods** to handle real-world constraints (occlusions, lighting, partial views), **improving robustness by 67%**.
- Resolved training plateaus through targeted **hyperparameter tuning & learning-rate scheduling**.

SKILLS

Technical experience & interests: Robotics Perception & Planning, Autonomous Systems, Robotic Simulation & Control, Reinforcement Learning, Embodied AI

Programming languages: Python, C/C++, HTML/CSS

Engineering tools: ROS, RViz, Gazebo, Linux, Git, MATLAB, LaTeX, Docker, SSH, Foxglove, PyTorch, YOLO, OpenCV, Py_trees, CloudCompare, CUDA

Languages: Vietnamese (Native), English (Fluent — IELTS 8.5)

Certifications & Activities: SAT: 1540/1600, AP Physics 1 & 2: 5/5, 1st Dan Taekwondo, Soccer, Piano & Drums, Chess